my sister's lethal recessives may be in the population at large, there is a disquietingly high chance that hers and mine are the same. If you do the sums, it turns out that, for every lethal recessive that I possess, if I mate with my sister one in eight of our offspring will be born dead or will die young. Incidentally, dying in adolescence is even more 'lethal', genetically speaking, than dying at birth: a stillborn child doesn't waste so much of the parents' vital time and energy. But, whichever way you look at it, close incest is not just mildly deleterious. It is potentially catastrophic. Selection for active incest-avoidance could be as strong as any selection pressure that has been measured in nature.

Anthropologists who object to Darwinian explanations of incest-avoidance perhaps do not realize what a strong Darwinian case they are opposing. Their arguments are sometimes so weak as to suggest desperate special pleading. They commonly say, for instance: 'If Darwinian selection had really built into us an instinctive revulsion against incest, we wouldn't need to forbid it. The taboo only grows up because people have incestuous lusts. So the rule against incest cannot have a "biological" function, it must be purely "social".' This objection is rather like the following: 'Cars don't need locks on the ignition switch because they have locks on the doors. Therefore ignition locks cannot be anti-theft devices; they must have some purely ritual significance!' Anthropologists are also fond of stressing that different cultures have different taboos, indeed different definitions of kinship. They seem to think that this, too, undermines Darwinian aspirations to explain incest-avoidance. But one might as well say that sexual desire cannot be a Darwinian adaptation because different cultures prefer to copulate in different positions. It seems to me highly plausible that incest-avoidance in humans, no less than in other animals, is the consequence of strong Darwinian selection.

Not only is it a bad thing to mate with those genetically too close to you. Too-distant outbreeding can also be bad because of genetic incompatibilities between different strains. Exactly where the ideal intermediate falls is not easy to predict. Should you mate with your first cousin? With your second or third cousin? Patrick Bateson has tried to ask Japanese quail where their own preferences lie along the spectrum. In an experimental setup called the Amsterdam Apparatus, birds were invited to choose among members of the opposite sex arrayed behind miniature shop-windows. They preferred first cousins over both full siblings and unrelated birds. Further experiments suggested that young quail learn the attributes of their clutch-companions, and then, later in life, tend to choose sexual partners that are quite like their clutch-mates but not too like them.

Quail, then, seem to avoid incest by their own internal lack of desire for those with whom they have grown up. Other animals do it by observing social laws, socially imposed rules of dispersal. Adolescent male lions, for instance, are driven out of the parental pride where female relatives remain to tempt them, and breed only if they manage to usurp another pride. In chimpanzee and gorilla societies it tends to be the young females who leave to seek mates in other bands. Both dispersal patterns, as well as the quail's system, are to be found among the various cultures of our own species.

p. 134 *Since [cuckoo hosts] are not in danger of being parasitized by members of their own species …*

This is probably true of most species of birds. Nevertheless, we should not be surprised to find some birds parasitizing nests of their own species. And the phenomenon is, indeed, being found in an increasing number of species. This is especially so now that new molecular techniques are coming in for establishing who is related to whom. Actually, the selfish gene theory might expect it to happen even more often than we so far know.

p. 136 *Kin selection in lions*